

Drone Design Specifications & Power-Weight Budget

2. Weight Budget (Upper Limits)

The dry weight of the drone structure and onboard electronics (excluding the main flight battery) is budgeted as follows:

Component Category	Component Description	Weight (g)
Flight Controller	PixHawk 6c Mini	~47 g
Companion Computer	Raspberry Pi	~50 g
Propulsion	4 x Holybro Motors (63 g each)	252 g
Primary Vision	6 x d500 Cameras (15g a piece)	270 g
Secondary Vision	2 x Arducam IMX219 (15g each)	~30 g
Receiver	RC Transmitter Receiver (RX)	~5 g
Communications	Telemetry Module	76 g
Video Transmission	VTX Module	5.5 g
Structure & Hardware	Miscellaneous (500 g Frame + 100 g extra hardware)	~600 g
Total Dry Weight (excl. batteries)	Excluding Battery	~1,335.5 g (1.34 kg)

3. Power Consumption Budget

Sustained Avionics & Payload Power:

- Pixhawk: 18 W
- Raspberry Pi: 15 W
- 6x d500 Cameras: 8.7 W
- 2x Arducam Units: 13.2 W
- RC Receiver (RX): 0.2 W
- Video Transmitter (VTX): 0.2 W
- Telemetry Module: 1.0 W

Motor Power Draw: Sustained: 880 W; Peak: 1,200+ W

Summary Power Metrics: Total Peak Consumption: ~1,250 W (Gives maximum current draw needed), Total Sustained Consumption: ~950 W (Used for endurance math).

4. Battery Energy & Endurance Calculations

A. Operational Target Baseline (10-Minute Mission)

1. Max Current = Peak Power / Voltage = 1250 W / 14.8 V = ~85 A
2. Total Energy = Sustained Power * Time = 950 W * (1/6 hr) = ~158 Wh
3. Minimum Calculated Capacity = (158 Wh / 14.8 V) * 1000 = ~11,000 mAh

B. Extended Mission Selection

- Selected Capacity: 15,000 mAh
- Battery Weight: 0.90 kg
- Flight Performance: Provides ~3.5 minutes of additional endurance, elevating total flight time to ~13.5 minutes.

5. Final Selection: Propulsion & ESC Specifications

Battery: Samsung INR 50S cells integrated into an Orange Robu.in ProRange battery pack.

ESC: T-Motor AT75A 2-6S ESC (Peak Current of 90 A, with a weight of about 80 g,).

Bonus Task

Alternate motor: Axisflying AE 3115 3-6S Motor (900 KV, Up to 3 kg of thrust per motor at 80% throttle vs. 0.75-1 by the Holybro at full throttle, might need stepup to meet 25V requiremet for these figures).